Sahaj Bhadja

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Summary

Software Test Engineer with 2+ years of experience in **autonomous-driving simulation**, **system validation**, test automation, failure analysis, coverage measurement, and reproducible evaluation across complex software systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Achieved 99%+ reliability across 5K+ monthly jobs through **validation, reconciliation, and error handling**, cloud storage, CI/CD releases, and production monitoring supporting dependable system-level test and operational workflows.
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant microservices** with explicit state transitions and failure isolation, preserving recoverable background processing under sustained production workloads reliably.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Maintained 90%+ scoring consistency across 10K+ submissions through **structured automated testing**, model switching, fallback paths, persistent results, and reviewer controls supporting reproducible evaluation and systematic failure analysis.
- Processed 1K+ files per batch using **multithreaded validation and fault isolation**, preserving responsive workflows while troubleshooting heterogeneous PDF, DOCX, and ZIP inputs under concurrent model execution reliably.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **reproducible autonomy-evaluation pipeline** over 100K+ driving logs with offline reinforcement learning and controlled experiments across diverse scenarios.
- Improved closed-loop planning stability by 15% through a **simulation-based validation model** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex scenarios reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.
- Improved API response times by 30% through **object-oriented services**, indexing, caching, pagination, and filtered payloads while debugging integrations and isolating performance bottlenecks across three production operating systems reliably.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst - Research Fellowship | *India*
- Achieved 96%+ classification accuracy across 650K+ URLs through a **reproducible test pipeline**, comparing Random Forest, CNN, RNN, LSTM, Bi-LSTM, and GRU models for malware and phishing detection reliably.
- Engineered 21 behavioral indicators covering URL length, IP presence, and suspicious keywords, translating noisy inputs into **validated classification features** that exposed failure patterns and improved analytical coverage at scale.

Skills

Languages: Python, C++, Java, SQL, JavaScript, TypeScript, Bash
Autonomy & Simulation: Autonomous Driving, Trajectory Planning, HD Maps, Reinforcement Learning, Simulation Testing, Scenario Testing, Closed-Loop Evaluation, Sensor Data, Planning Systems
Testing & Safety: Test Planning, Test Execution, Unit Testing, Integration Testing, Automated Testing, Regression Testing, System Testing, Test Coverage, Validation, Verification, Failure Analysis, Root-Cause Analysis
Data & Analysis: Data Pipelines, Data Validation, Feature Engineering, Model Evaluation, Performance Metrics, Classification, PyTorch, Pandas, NumPy, Matplotlib
Tools & Delivery: Linux, Git, GitHub, Jenkins, Docker, CI/CD, PyTest, Mockito, Debugging, Monitoring
Engineering Practices: Requirements Analysis, Technical Documentation, Test Reports, Issue Tracking, Fault Isolation, Performance Optimization, Cross-Functional Collaboration, Technical Communication, Software Development Lifecycle

Projects

Autonomy Scenario Validation Harness - Design Concept
- Designed a **Python/C++ validation-harness concept** to execute deterministic driving scenarios, inject sensor and planning faults, capture traces, compare safety metrics against thresholds, and produce reproducible failure reports for developers.

Simulation Coverage and Failure Analyzer - Design Concept
- Designed a **simulation-analysis workflow concept** aggregating scenario tags, autonomy-function results, failure clusters, root-cause evidence, and regression histories into coverage dashboards and prioritized test-plan recommendations for systems teams.

CPU-Scheduling Simulator | Live
- Built a **CPU-scheduling simulator** implementing FCFS, SJF, SRTF, Priority, and Round Robin algorithms with configurable workloads, timeline visualization, comparative metrics, and deterministic validation across scheduling scenarios reliably.